PROJECT PRESENTATION

COLLEGE ATTENDANCE MANAGEMENT SYSTEM

A simple digital solution to record, monitor, and manage student attendance.

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01 Project Overview

What the system does and why it is useful

Digital attendance

Teachers can mark attendance for each class and date.

Centralized records

Attendance information is stored in one organized system.

Quick monitoring

Students and staff can review attendance summaries.

Main idea

Replace scattered paper registers and manual calculations with a reliable, easy-to-use digital workflow.

02 Problem Statement

Challenges in a manual attendance process

Time-consuming

Teachers spend class time writing names and marking registers.

Calculation errors

Manual totals can lead to mistakes in attendance percentages.

Difficult tracking

Finding a student's attendance history takes extra effort.

Record handling

Paper sheets can be misplaced, damaged, or hard to share.

03 Project Objectives

What we aim to achieve

1. Reduce the time required to record attendance.
2. Maintain accurate and organized attendance records.
3. Calculate attendance percentages automatically.
4. Help staff identify students with low attendance.
5. Provide clear summaries for students and administrators.

04 Users & Roles

Different users have different permissions

Administrator

- Manage student and staff accounts
- Maintain course and class details
- View overall attendance reports

Faculty / Teacher

- Select class, subject, and date
- Mark students present or absent
- Edit records when authorized

Student

- View personal attendance summary
- Check attendance percentage
- Receive low-attendance alerts (if enabled)

Access should be role-based so users can only perform actions they are allowed to use.

05 Key Features

Core functions of the proposed system

Login & roles

Secure sign-in for admin, faculty, and students.

Class management

Organize departments, courses, subjects, and sections.

Attendance marking

Record present / absent for a selected class and date.

Automatic reports

Calculate totals and attendance percentages.

Search & filters

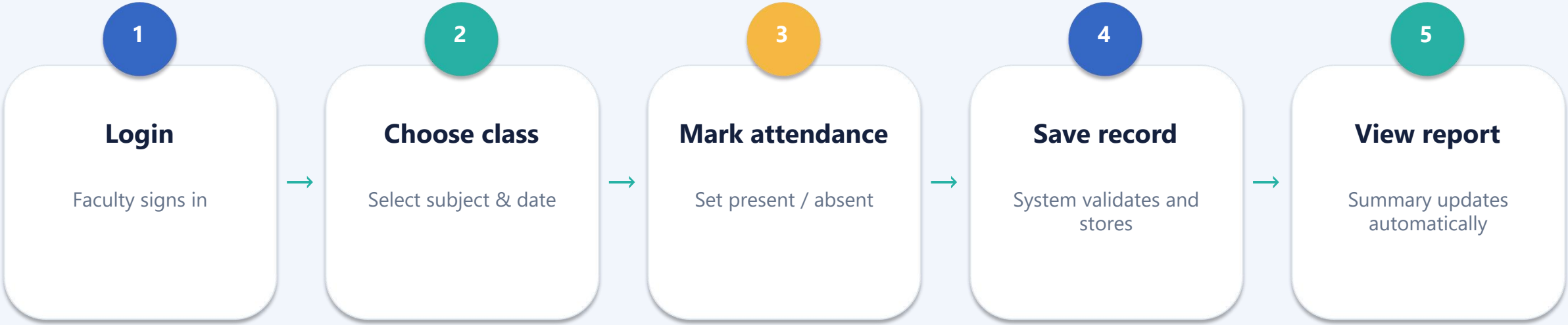
Find records by student, class, subject, or date.

Alerts

Flag students below a configurable attendance threshold.

06 System Workflow

A typical attendance session



Validation examples: prevent duplicate entries, require a class/date, and confirm the record is saved.

07 System Requirements

Functional and non-functional requirements

Functional requirements

- User login and role permissions
- Add / update student details
- Mark and save attendance
- Search and generate reports
- Calculate attendance percentage

Non-functional requirements

- Security: protect student information
- Usability: simple and clear interface
- Reliability: preserve saved records
- Performance: display reports quickly
- Maintainability: easy to update

Requirements should be clear, measurable, testable, and unambiguous.

08 Proposed Technology Stack

Choose tools that match the team's skills and project scope

Frontend

HTML, CSS, JavaScript — Screens, forms, and dashboards

Backend

Python / Java / Node.js — Business rules and access control

Database

MySQL / SQLite — Student, class, and attendance records

Version control

Git + GitHub — Track changes and collaborate

Note: This is a proposed stack; replace it with the tools actually used by your team.

09 Basic Data Design

Example entities for storing attendance

STUDENT

- student_id (PK)
- name
- roll_number
- section

→

CLASS / SUBJECT

- class_id (PK)
- subject_name
- faculty_id
- section

→

ATTENDANCE

- attendance_id (PK)
- student_id (FK)
- class_id (FK)
- date, status

PK = Primary Key | FK = Foreign Key | Attendance links each student to a class and date.

10 Security & Testing

Protect records and verify the main workflows

Security considerations

- Role-based access control
- Strong password handling
- Validate user inputs
- Limit access to personal records
- Back up the database regularly

Test cases

- Valid login succeeds; invalid login fails
- Attendance saves for correct class/date
- Duplicate entries are blocked
- Percentage matches saved records
- Unauthorized users cannot edit data

11 Benefits & Limitations

Expected impact and points to manage

Benefits

- Faster attendance recording
- Fewer manual calculation errors
- Easy access to attendance history
- Better visibility of low attendance
- Less paper-based record keeping

Limitations to address

- Requires device and power access
- Network may be needed for shared access
- Incorrect entries still need correction
- Student data must be kept private

12 Future Enhancements

Features that can be added in later versions

QR-based check-in

Speed up classroom attendance with controlled QR codes.

Parent notifications

Send alerts when attendance falls below the set limit.

Mobile-friendly app

Let students and faculty use phones easily.

Analytics dashboard

Show attendance trends by class and subject.

CONCLUSION

A smarter way to manage college attendance.

The system organizes attendance records, reduces routine work, and helps the college monitor attendance more effectively.

THANK YOU

Questions?